

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$x = 49$

$y = \sqrt{x} + 9$

The graphs of the given equations intersect at the point (x, y) in the xy -plane. What is the value of y ?

Answer

- A. 16
- B. 40
- C. 81
- D. 130

Correct Answer: A

Rationale

Choice A is correct. It's given that the graphs of the given equations intersect at the point (x, y) in the xy -plane. It follows that (x, y) represents a solution to the system consisting of the given equations. The first equation given is $x = 49$. Substituting 49 for x in the second equation given, $y = \sqrt{x} + 9$, yields $y = \sqrt{49} + 9$, which is equivalent to $y = 7 + 9$, or $y = 16$. It follows that the graphs of the given equations intersect at the point $(49, 16)$. Therefore, the value of y is 16.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.